

MBAI 5310G – Final Project One-Page Concept

Project Title: Predicting Gen Z Mobile Banking Engagement Using Machine Learning

Research Topic: This project explores the factors influencing Generation Z engagement with mobile banking applications, focusing on the moderating role of perceived trust in perceived transparency, perceived reward, and technological readiness.

Problem Statement: Mobile banking adoption among Gen Z users depends on multiple psychological and technological factors. This project aims to examine how trust influences the relationship between transparency, rewards, technological readiness, and user engagement.

Project Objectives: Analyze factors affecting Gen Z mobile banking engagement. Build machine learning models to predict engagement levels. Compare the performance of classification models. Interpret business implications and responsible AI considerations.

Dataset: Survey-based or publicly available mobile banking datasets containing variables such as perceived trust, transparency, rewards, technological readiness, and engagement level.

Proposed Machine Learning Models: Logistic Regression Decision Tree Random Forest Tools and Technologies: Python, Jupyter Notebook/Google Colab, Pandas, Scikit-learn, and GitHub for reproducible workflow and documentation.

Expected Outcomes: The project is expected to identify the most important factors influencing Gen Z engagement in mobile banking and evaluate how trust moderates these relationships.

Responsible AI Reflection: The project will address data privacy, ethical use of financial data, bias in prediction systems, and the importance of transparency and human oversight in AI-driven financial applications.

Course Alignment: This project aligns with MBAI 5310G course expectations by incorporating machine learning implementation, model evaluation, GitHub documentation, reproducible workflow, business interpretation, and responsible AI reflection.